A Counterexample to General Convergence of Double Q-Learning with Linear Function Approximation

1 Claim

Double Q-learning is convergent in the tabular case under standard assumptions. This proof shows that the same is not true for arbitrary linear function approximation.

Theorem 1. *There exists a finite discounted Markov decision process, a stationary behavior policy, a bounded linear feature map, and a Robbins–Monro stepsize sequence such that symmetric Double Q-learning with linear function approximation is unbounded almost surely from any strictly positive initialization. This holds even though the optimal action-value function is identically zero and is exactly representable by the feature map.*

The example has one state, two actions, zero rewards, and one scalar feature.

2 Preliminaries

Definition 1 (Stochastic process)**.** *A stochastic process is a collection of random variables $\{Z_n\}_{n\geq 0}$ defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$, each taking values in a measurable space $\mathcal{E}$. The index n typically represents discrete time, so Z_n is the random state of the process at time n.*

Definition 2 (Markov Decision Process)**.** *A finite discounted Markov decision process is a tuple $(\mathcal{S}, \mathcal{U}, P, r, \gamma)$ where $\mathcal{S}$ is a finite state space, $\mathcal{U}$ is a finite action space, $P(x' \mid x, u)$ is a transition kernel giving the probability of moving to state $x' \in \mathcal{S}$ from state $x \in \mathcal{S}$ under action $u \in \mathcal{U}$, $r\colon \mathcal{S} \times \mathcal{U} \to \mathbb{R}$ is a bounded reward function, and $\gamma \in [0, 1)$ is the discount factor.*

Definition 3 (State process and input process)**.** *The state process $\{X_n\}_{n\geq 0}$ is an $\mathcal{S}$-valued stochastic process satisfying $\mathbb{P}(X_{n+1} = x' \mid X_0, U_0, \dots, X_n, U_n) = P(x' \mid X_n, U_n)$. The input process $\{U_n\}_{n\geq 0}$ is a $\mathcal{U}$-valued process representing the sequence of actions taken by the agent. Under a stationary behavior policy μ, the input process satisfies $U_n \sim \mu(\cdot \mid X_n)$ for each n.*

The action-value function under a policy μ is $Q^\mu(x, u) = \mathbb{E}\big[\sum_{k=0}^{\infty} \gamma^k\, r(X_{n+k}, U_{n+k}) \mid X_n = x,\, U_n = u\big]$, and the optimal action-value function is $Q^*(x, u) = \max_\mu Q^\mu(x, u)$. Q-learning and its variants seek to estimate Q^* from observed transitions.

In the remainder of this paper, both $\mathcal{S}$ and $\mathcal{U}$ are assumed to be finite.

3 The MDP, features, and behavior policy

Consider the discounted MDP with one state s, two actions a_1, a_2, deterministic self-transition, zero rewards, and discount factor $\gamma = 0.9$. Thus $P(s \mid s, a) = 1$ and $r(s, a) = 0$ for both actions $a \in \{a_1, a_2\}$. Since all rewards are zero,

$$Q^*(s, a_1) = Q^*(s, a_2) = 0. \tag{1}$$

Use one-dimensional linear features $\phi(s,a_1)=1$ and $\phi(s,a_2)=2$. The two Double-Q estimators are parameterized by scalars θ_A,θ_B:

$$Q_A(s,a)=\theta_A\phi(s,a), \qquad Q_B(s,a)=\theta_B\phi(s,a). \tag{2}$$

The optimal function $Q^*\equiv 0$ is exactly representable by the parameter value $\theta_A=\theta_B=0$.

Let the stationary behavior policy be $\mu(a_1 \mid s)=0.9$ and $\mu(a_2\mid s)=0.1$. At time n, sample an action from μ, and write $X_{n+1}=\phi(s,A_{n+1})\in\{1,2\}$. Hence

$$\mathbb{P}(X_{n+1}=1)=0.9, \qquad \mathbb{P}(X_{n+1}=2)=0.1. \tag{3}$$

Independently of the sampled action and of the past, choose estimator A or estimator B for update with probability 1/2 each. Use the global stepsize sequence

$$\alpha_n=\frac{a}{n+1}, \qquad 0<a\le\frac{1}{16}. \tag{4}$$

Then $\sum_{n=0}^{\infty}\alpha_n=\infty$ and $\sum_{n=0}^{\infty}\alpha_n^2<\infty$. Initialize with arbitrary strictly positive parameters: $\theta_{A,0}>0$ and $\theta_{B,0}>0$.

4 The Double Q-learning updates

We use the standard symmetric Double Q-learning update. If estimator A is selected at time n, then

$$\theta_{A,n+1}=\theta_{A,n}+\alpha_n\phi(s,A_{n+1})\big[\gamma\,Q_B\big(s,\arg\max_{a'}Q_A(s,a')\big)-Q_A(s,A_{n+1})\big], \tag{5}$$

while θ_B is unchanged. If estimator B is selected, the roles are reversed:

$$\theta_{B,n+1}=\theta_{B,n}+\alpha_n\phi(s,A_{n+1})\big[\gamma\,Q_A\big(s,\arg\max_{a'}Q_B(s,a')\big)-Q_B(s,A_{n+1})\big], \tag{6}$$

while θ_A is unchanged. Because the state never changes and the reward is zero, these are the full updates.

5 Positivity is invariant

We first show that the positive quadrant is invariant. This will imply that the greedy action is fixed throughout the entire trajectory.

Suppose $\theta_A>0$. Then $Q_A(s,a_2)=2\theta_A>\theta_A=Q_A(s,a_1)$, so the greedy action under Q_A is uniquely a_2. Similarly, if $\theta_B>0$, the greedy action under Q_B is uniquely a_2. Thus no tie-breaking convention is needed as long as the parameters remain strictly positive.

Assume $\theta_A,\theta_B>0$, and let $X\in\{1,2\}$ be the sampled feature. If estimator A is updated, then the target action is a_2, whose feature is 2. Therefore

$$\theta_A^+=\theta_A+\alpha X(\gamma\cdot 2\theta_B-X\theta_A)=(1-\alpha X^2)\theta_A+1.8\,\alpha X\,\theta_B. \tag{7}$$

Because $X\le 2$ and $\alpha_n\le 1/16$, we have $1-\alpha_nX^2\ge 1-4\alpha_n\ge 3/4>0$. Thus $\theta_A^+>0$. The same argument applies when estimator B is updated:

$$\theta_B^+=(1-\alpha X^2)\theta_B+1.8\,\alpha X\,\theta_A>0. \tag{8}$$

Therefore, since $\theta_{A,0}>0$ and $\theta_{B,0}>0$, we have $\theta_{A,n}>0$ and $\theta_{B,n}>0$ for every n. Consequently, the greedy action is always a_2 for both estimators. Along this trajectory, the nonlinear maximization in the Double-Q target reduces to a fixed linear target.

6 Dynamics of the parameter sum

Define $\Sigma_n = \theta_{A,n} + \theta_{B,n}$. By positivity invariance, $\Sigma_n > 0$ for all n. We will prove that $\Sigma_n \to \infty$ almost surely. This implies that the Double-Q parameter vector is unbounded and therefore cannot converge to $(0, 0)$, which represents Q^*.

Let $r_n = \theta_{A,n}/\Sigma_n \in (0, 1)$. If estimator A is updated at time n, then

$$\Sigma_{n+1} = \Sigma_n + \alpha_n X_{n+1}(1.8\,\theta_{B,n} - X_{n+1}\,\theta_{A,n}). \tag{9}$$

Dividing by Σ_n, this becomes $\Sigma_{n+1} = \Sigma_n(1 + \alpha_n Y_{n+1})$, where, in the case of an A-update,

$$Y_{n+1} = X_{n+1}\big(1.8(1 - r_n) - X_{n+1}\,r_n\big). \tag{10}$$

If estimator B is updated, then similarly $\Sigma_{n+1} = \Sigma_n(1 + \alpha_n Y_{n+1})$, where

$$Y_{n+1} = X_{n+1}\big(1.8\,r_n - X_{n+1}(1 - r_n)\big). \tag{11}$$

For both possible updates, because $r_n \in (0, 1)$ and $X_{n+1} \in \{1, 2\}$, we have $-4 \le Y_{n+1} \le 3.6$. Therefore $|\alpha_n Y_{n+1}| \le 1/4$ and $1 + \alpha_n Y_{n+1} \ge 3/4 > 0$. Thus the logarithm of Σ_n is well defined for every n.

7 Positive drift of the logarithm

Let $\mathcal{F}_n$ be the history up to time n, including $\theta_{A,n}$ and $\theta_{B,n}$, but not the fresh action sample or estimator choice at time n. Conditional on $\mathcal{F}_n$, the ratio r_n is fixed.

First average over the independent choice of which estimator is updated. For a fixed sampled feature X,

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n, X_{n+1} = X] &= \tfrac{1}{2}\,X\big(1.8(1 - r_n) - X\,r_n\big) + \tfrac{1}{2}\,X\big(1.8\,r_n - X(1 - r_n)\big)\\ &= \tfrac{1}{2}\,X(1.8 - X).\end{aligned} \tag{12}$$

Now average over the behavior policy:

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] &= 0.9 \cdot \tfrac{1}{2} \cdot 1 \cdot (1.8 - 1) + 0.1 \cdot \tfrac{1}{2} \cdot 2 \cdot (1.8 - 2)\\ &= 0.9 \cdot 0.4 + 0.1 \cdot (-0.2)\\ &= 0.36 - 0.02 = 0.34.\end{aligned} \tag{13}$$

Thus Y_{n+1} has a strictly positive conditional mean, uniformly over the current parameters.

Define $L_n = \log \Sigma_n$. Since $\Sigma_{n+1} = \Sigma_n(1 + \alpha_n Y_{n+1})$, we have $L_{n+1} - L_n = \log(1 + \alpha_n Y_{n+1})$. For $|t| \le 1/4$, the elementary inequality $\log(1 + t) \ge t - t^2$ holds. Applying this with $t = \alpha_n Y_{n+1}$, and using $|Y_{n+1}| \le 4$, gives

$$\mathbb{E}[L_{n+1} - L_n \mid \mathcal{F}_n] \ge \alpha_n\,\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] - \alpha_n^2\,\mathbb{E}[Y_{n+1}^2 \mid \mathcal{F}_n] \ge 0.34\,\alpha_n - 16\,\alpha_n^2. \tag{14}$$

Because $\sum_{n=0}^{\infty} \alpha_n = \infty$ and $\sum_{n=0}^{\infty} \alpha_n^2 < \infty$, we have $\sum_{n=0}^{N-1}(0.34\,\alpha_n - 16\,\alpha_n^2) \to +\infty$. So the predictable drift of L_n has a deterministic lower bound whose partial sums diverge to $+\infty$.

8 Martingale fluctuations are negligible

Let $\Delta L_{n+1} = L_{n+1} - L_n = \log(1 + \alpha_n Y_{n+1})$, and define the martingale difference

$$M_{n+1} = \Delta L_{n+1} - \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n]. \tag{15}$$

For $|t| \le 1/4$, there is a universal constant K such that $|\log(1+t)| \le K|t|$. For example, one may take $K = 2$. Therefore $|\Delta L_{n+1}| \le 2\alpha_n |Y_{n+1}| \le 8\alpha_n$. Consequently,

$$\big|\mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n]\big| \le \mathbb{E}\big[|\Delta L_{n+1}| \mid \mathcal{F}_n\big] \le 8\alpha_n. \tag{16}$$

Hence $|M_{n+1}| \le |\Delta L_{n+1}| + |\mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n]| \le 16\alpha_n$. It follows that $\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] \le 256\,\alpha_n^2$. Since $\sum_n \alpha_n^2 < \infty$,

$$\sum_{n=0}^{\infty} \mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] < \infty. \tag{17}$$

By the martingale convergence theorem for square-integrable martingales with summable conditional variances, the martingale partial sums $\sum_{n=0}^{N-1} M_{n+1}$ converge almost surely to a finite random limit.

9 Almost-sure divergence

Using the decomposition

$$L_N = L_0 + \sum_{n=0}^{N-1} \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n] + \sum_{n=0}^{N-1} M_{n+1}, \tag{18}$$

and the lower bound from Section 6,

$$\sum_{n=0}^{N-1} \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n] \ge \sum_{n=0}^{N-1} (0.34\,\alpha_n - 16\,\alpha_n^2) \to +\infty. \tag{19}$$

The martingale term converges almost surely to a finite limit. Therefore $L_N \to +\infty$ almost surely. Since $L_N = \log \Sigma_N$, this implies $\Sigma_N \to \infty$ almost surely. Since both parameters remain positive,

$$\max\{\theta_{A,N},\, \theta_{B,N}\} \ge \frac{\Sigma_N}{2} \to \infty. \tag{20}$$

Thus the parameter vector is unbounded almost surely. In particular, the Double-Q iterates cannot converge to the representable optimum $(\theta_A, \theta_B) = (0, 0)$. This proves the theorem. $\square$

10 Relation to the expected linear dynamics

This section is not needed for the almost-sure proof, but it explains the source of the instability. In the positive quadrant, the greedy action is always a_2. Therefore the expected update is linear. Let $C = \mathbb{E}[X^2]$ and $D = \gamma\,\mathbb{E}[2X]$. In this example, $C = 0.9 \cdot 1^2 + 0.1 \cdot 2^2 = 1.3$ and $D = 0.9(0.9 \cdot 2 \cdot 1 + 0.1 \cdot 2 \cdot 2) = 1.98$. The expected Double-Q update has the form

$$\mathbb{E}[\Delta\theta_A] = \frac{\alpha}{2}(D\,\theta_B - C\,\theta_A), \qquad \mathbb{E}[\Delta\theta_B] = \frac{\alpha}{2}(D\,\theta_A - C\,\theta_B). \tag{21}$$

Equivalently,

$$\mathbb{E}\begin{bmatrix}\Delta\theta_A \\ \Delta\theta_B\end{bmatrix} = \frac{\alpha}{2}\begin{bmatrix}-C & D \\ D & -C\end{bmatrix}\begin{bmatrix}\theta_A \\ \theta_B\end{bmatrix}. \tag{22}$$

Introduce the average and difference coordinates $u = (\theta_A + \theta_B)/2$ and $v = (\theta_A - \theta_B)/2$. The corresponding mean ODE decouples as

$$\dot{u} = \tfrac{1}{2}(D - C)\, u, \qquad \dot{v} = -\tfrac{1}{2}(C + D)\, v. \tag{23}$$

Since $D - C = 1.98 - 1.3 = 0.68 > 0$, we obtain $\dot{u} = 0.34\, u$. Thus the average mode is unstable. The almost-sure argument above turns this mean-dynamics instability into a genuine stochastic nonconvergence result.

11 What the counterexample shows and does not show

This counterexample shows that there cannot be a general convergence theorem for Double Q-learning with arbitrary linear function approximation under off-policy sampling and bootstrapping.

It does not show that Double Q-learning always diverges. Double Q-learning is convergent in tabular settings under the usual assumptions, and it may converge in restricted linear settings with additional structure. The point is that Double Q-learning does not, by itself, remove the instability caused by the combination of bootstrapping, off-policy sampling, and function approximation.

The example is deliberately minimal: one state, two actions, zero rewards, one scalar feature, and a fixed behavior policy. Despite this simplicity, the Double-Q parameter norm diverges almost surely from every strictly positive initialization.